

Positive-real asymptotic inversion

Power-logarithmic series, exact irrational exponents,
and a Wolfram Language implementation

Technical article and implementation
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Abstract

We construct the positive-real inverse branch at zero of $f(x) = ax^\alpha(1 + \sum_j x^{\delta_j} P_j(\log x))$, where $a, \alpha, \delta_j > 0$ and the P_j are real polynomials. A universal Euler–Lagrange formula produces every coefficient without a closed-form inverse, rationalizing exponents, or treating logarithms as constants under differentiation. A finite positive exponent semigroup makes weighted truncation an exact finite computation. A multivariate analytic implicit-function argument proves convergence for the finite input class; this is stronger than a merely formal expansion. We prove a first-omitted power theorem that preserves the logarithmic factor in the error term, and explain how inverse residuals and separately known input errors are transported. The two motivating examples, $x + x^2(1 + \log x)$ and $x + x^{\sqrt{2}}$, receive explicit formulas, branch analysis, and numerical checks. The accompanying Wolfram Language package implements sparse coefficient generation, exact exponent comparison, conservative remainder metadata, and an independent binomial/logarithm composition checker. Its scope is deliberately a precise power-logarithmic class, rather than an unspecified universal transseries solver.

Deliverables and execution status. The archive contains this article and its source, the package, 35 Wolfram regression tests, runnable examples, and independent validation scripts. The mathematical algorithms passed 58 independently executed SymPy/mpmath checks. A Wolfram kernel was unavailable in this session; the package and its MUnit suite have not been executed in a proprietary Wolfram runtime. Section 12 separates these statements in detail.

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1 The question is about a germ, not just an inverse symbol

The supplied question [1] asks for the real inverse of

$$f_1(x) = x + x^2(1 + \log x), \quad x > 0, \quad (1.1)$$

near the image point zero. A second example asks for the inverse of

$$f_2(x) = x + x^{\sqrt{2}}, \quad x \geq 0. \quad (1.2)$$

The archived evaluation of an unqualified `InverseFunction` produces an expansion with a nonreal constant term in the first example. This does not identify the desired germ. The specification needed here is

$$f(g(y)) = y, \quad y \downarrow 0, \quad g(y) > 0, \quad g(y) \sim y. \quad (1.3)$$

The target point alone does not specify a preimage or a branch. In particular, putting a reality condition around a symbolic expression is not the same as constructing the local inverse germ with a stated asymptotic leading term. We analyze the archived output, not the behavior of an untested current Wolfram release.

1.1 The first example actually has a unique positive inverse globally

Differentiation gives

$$f_1'(x) = 1 + x(3 + 2 \log x).$$

The function $x(3 + 2 \log x)$ has its global minimum at $x = e^{-5/2}$, where its value is $-2e^{-5/2}$. Consequently

$$f_1'(x) \geq 1 - 2e^{-5/2} > 0 \quad (x > 0). \quad (1.4)$$

Moreover $f_1(x) \rightarrow 0$ as $x \downarrow 0$, and $f_1(x) \rightarrow \infty$ as $x \rightarrow \infty$. Thus it is a bijection of the positive half-line onto itself. There is no ambiguity about its positive-real inverse. The difficulty is representing its nonanalytic endpoint germ.

For comparison, a nonzero complex zero of the analytic continuation satisfies $x(1 + \log x) = -1$. Writing $w = 1 + \log x$ gives $we^w = -e$, hence

$$x = \exp(W_k(-e) - 1) = -\frac{1}{W_k(-e)}$$

where the chosen logarithm and Lambert branch must be compatible. The nonreal constant displayed in the archived question is of this form. An ordinary analytic inverse expansion there belongs to a different preimage of zero, not to the positive endpoint branch.

For f_2 , the derivative $1 + \sqrt{2}x^{\sqrt{2}-1}$ is positive for $x > 0$. This also gives a unique positive inverse, globally.

1.2 Two independent representation issues

The logarithmic example does not have an ordinary Taylor expansion at zero. For instance, the third derivative of $x^2 \log x$ is $2/x$. The second example introduces the irrational exponent gap $\sqrt{2} - 1$; no integer ramification turns all its inverse exponents into integers.

The documented `SeriesData` representation uses one lattice of powers $(n_{\min} + j)/\text{den}$, and `InverseSeries` operates on suitably structured objects of that kind [7, 8]. This is not the representation used below. A sparse list of exact real exponents with polynomial-logarithmic coefficients expresses both examples directly. Wolfram's `Asymptotic` documentation allows general inferred scales, but that does not remove the need to specify the inverse germ or establish the appropriate coefficient algebra [9].

1.3 How this relates to the supplied transseries material

The ProveIt material [2] distinguishes support conditions, formal summability, compositional inversion, logarithmic degree growth, and analytic remainder claims. Its chapter on the Lagrange–Bürmann formula at infinity is particularly pertinent. We use the same separation, but work at 0^+ , allow finitely many exact irrational exponent generators, normalize an arbitrary positive leading monomial, and prove an analytic realization for the finite input class.

The broader problem belongs to the theory of transseries and multiseries. Edgar’s introduction and van der Hoeven’s monograph provide background; Salvy and Shackell specifically study algorithms for multiseries expansions of inverse exp-log functions [4, 5, 6]. The package here is a specialized implementation with a narrower, explicit contract. Its proofs below do not assume a black-box theorem saying that every imaginable transseries can be substituted or inverted.

2 The input class and the real branch

Fix real constants $a > 0$, $\alpha > 0$, and a finite collection of positive numbers $\delta_1, \dots, \delta_r$. Consider

$$f(x) = ax^\alpha(1 + u(x)), \quad u(x) = \sum_{j=1}^r x^{\delta_j} P_j(\log x), \quad P_j \in \mathbb{R}[L]. \quad (2.1)$$

Combine equal δ_j and discard zero polynomials. The coefficients may depend on fixed real parameters; all assertions are then for parameter values satisfying the stated hypotheses, not automatically uniform as a parameter approaches a singular boundary.

Throughout, real powers mean $x^q = \exp(q \log x)$ for $x > 0$. Set

$$t = (y/a)^{1/\alpha}, \quad L = \log t, \quad \mathcal{E} = t \frac{d}{dt}. \quad (2.2)$$

The required branch is the one satisfying $g(y)/t \rightarrow 1$. The symbol L is an independent polynomial coordinate during coefficient arithmetic, but its differential rule is never forgotten:

$$\mathcal{E}[t^\mu Q(L)] = t^\mu (\mu + \partial_L) Q(L). \quad (2.3)$$

Theorem 2.1 (Existence and selection of the positive branch). *For a nonzero finite perturbation in (2.1), let $\delta_* = \min_j \delta_j$ and $d = \max_j \deg P_j$. For sufficiently small $\varepsilon > 0$, f is positive and strictly increasing on $(0, \varepsilon)$, extends continuously by $f(0) = 0$, and has a unique positive inverse there. That inverse satisfies*

$$g(y) \sim (y/a)^{1/\alpha}, \quad g'(y) \sim \frac{(y/a)^{1/\alpha}}{\alpha y}.$$

Proof. Every fixed power of $|\log x|$ is dominated by every positive power of x^{-1} . Therefore

$$u(x) = \mathcal{O}\left(x^{\delta_*}(1 + |\log x|^d)\right) = o(1), \quad xu'(x) = o(1).$$

The derivative is

$$f'(x) = ax^{\alpha-1} [\alpha(1 + u(x)) + xu'(x)] \sim a\alpha x^{\alpha-1} > 0.$$

This proves local positivity and monotonicity. From $y = ag(y)^\alpha(1 + u(g(y)))$ we obtain $g(y)/t \rightarrow 1$. Finally $g' = 1/f'(g)$ yields the derivative equivalent. The argument does not require a finite nonzero derivative of f at the endpoint; thus $\alpha < 1$ and $\alpha > 1$ are both allowed. $\square$

3 The coefficient algebra and why truncation is finite

3.1 A positive semigroup, not a rational exponent lattice

Let

$$S = \{m_1\delta_1 + \cdots + m_r\delta_r : (m_1, \dots, m_r) \in \mathbb{N}^r\}. \quad (3.1)$$

For every finite B , only finitely many multiindices can satisfy $\sum m_j\delta_j \leq B$, because $m_j \leq B/\delta_j$. Thus S is locally finite. This remains true when some ratios δ_i/δ_j are irrational. The additive *group* they generate may be dense; the nonnegative *semigroup* relevant here is locally finite. Confusing these two statements can lead to an incorrect objection to finite coefficient extraction.

Define the formal algebra

$$\mathcal{A}_\Delta = \left\{ \sum_{\mu \in S} t^\mu Q_\mu(L) : Q_\mu \in \mathbb{R}[L] \right\}, \quad \text{val } A = \min\{\mu : Q_\mu \neq 0\}.$$

Each bounded power interval contains only finitely many polynomial blocks. Multiplication is the finite convolution

$$[t^\mu](AB) = \sum_{\lambda+\nu=\mu} A_\lambda(L)B_\nu(L).$$

The Euler derivation preserves power valuation and does not increase logarithmic degree. A fixed coefficient below a power cutoff is therefore computed with finitely many exact polynomial operations.

3.2 Composition near the identity

For v of positive valuation, the expressions

$$(1+v)^q = \sum_{n \geq 0} \binom{q}{n} v^n, \quad \log(1+v) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} v^n \quad (3.2)$$

are formally summable. Substitution into a block uses

$$(t(1+v))^\delta P(\log(t(1+v))) = t^\delta (1+v)^\delta P(L + \log(1+v)). \quad (3.3)$$

This gives a complete rule for composition within the class. In particular, it differentiates and shifts logarithms correctly; replacing $\log t$ by an unrelated constant before reversing a series would not implement this rule.

Proposition 3.1 (Formal existence and uniqueness). *There is a unique $s \in 1 + \mathcal{A}_{\Delta, >0}$ satisfying*

$$s^\alpha (1 + u(ts)) = 1. \quad (3.4)$$

Its support is contained in S , and the formal inverse is $g = ts$.

Proof. Rewrite the equation as $s = (1 + u(ts))^{-1/\alpha}$, with the binomial branch having constant term one. If $s_1 - s_2$ has valuation $\eta > 0$, formula (3.3) shows that $u(ts_1) - u(ts_2)$ has valuation at least $\eta + \delta_*$. Applying the unit power preserves that gain. Starting from $s_0 = 1$, iteration therefore stabilizes every bounded set of coefficients after finitely many steps. The same valuation gain proves uniqueness: a nonzero least-weight difference of two solutions would have to gain positive weight while remaining equal to itself. All operations preserve the semigroup. $\square$

This proposition also applies to locally finite infinite formal inputs with a strictly positive least correction weight. It says nothing by itself about convergence or the analytic size of a remainder. Those require the separate arguments in Sections 5 and 6.

4 A universal Euler–Lagrange inversion formula

Theorem 4.1 (Euler–Lagrange formula). *For (2.1), the normalized inverse is*

$$g(y) = t \left[1 + \sum_{n \geq 1} \frac{(-1)^n}{n! \alpha^n} \prod_{j=1}^{n-1} (\mathcal{E} + 1 + j\alpha) u(t)^n \right]. \quad (4.1)$$

An empty operator product is the identity. The formula is valid formally; for the finite input class it also converges for sufficiently small positive y , as proved in Theorem 5.1.

4.1 Derivation without assuming analyticity at zero

Classical Lagrange inversion is an analytic local theorem at a regular point [3]. We apply it in an auxiliary perturbation parameter, around a positive point, not at the singular endpoint.

Introduce λ and consider

$$x^\alpha (1 + \lambda u(x)) = t^\alpha.$$

For fixed $t > 0$, put $w = t^\alpha$, $z = x^\alpha$, and

$$R(z) = z u(z^{1/\alpha}), \quad H(z) = z^{1/\alpha}.$$

All functions are analytic near the positive point $z = w$, with consistent local branches. The equation becomes $z + \lambda R(z) = w$. The parameter form of Lagrange–Bürmann gives

$$H(z(\lambda)) = H(w) + \sum_{n \geq 1} \frac{(-\lambda)^n}{n!} \left(\frac{d}{dw} \right)^{n-1} [H'(w) R(w)^n]. \quad (4.2)$$

For completeness, the coefficient identity follows by the residue formula for $H(z(\lambda))$, changing variables from the inverse to $z + \lambda R(z) - w$, and integrating the derivative of the logarithm once by parts. At order λ^n the residue is

$$\frac{(-1)^n}{n} \operatorname{Res}_{z=w} \frac{H'(z) R(z)^n}{(z-w)^n} = \frac{(-1)^n}{n!} \left(\frac{d}{dw} \right)^{n-1} [H'(w) R(w)^n].$$

Equivalently, this is the usual formal-parameter reversion identity; no convergence of a Taylor series in t about zero has been invoked.

Here

$$H'(w) R(w)^n = \frac{1}{\alpha} w^{n-1+1/\alpha} u(w^{1/\alpha})^n.$$

Repeated differentiation, followed by $w = t^\alpha$, yields

$$\left(\frac{d}{dw} \right)^{n-1} [w^{n-1+1/\alpha} A(w^{1/\alpha})] = \frac{t}{\alpha^{n-1}} \prod_{j=1}^{n-1} (\mathcal{E} + 1 + j\alpha) A(t). \quad (4.3)$$

For a monomial $A(t) = t^\mu$, both sides have the same product of $n-1$ linear factors. The identity for general differentiable A follows by iterating $d/dw = (\alpha t^{\alpha-1})^{-1} d/dt$. Substitution into (4.2) gives (4.1). Formal summability follows because its n -th correction has relative valuation at least $n\delta_*$. Setting $\lambda = 1$ analytically is justified separately below.

4.2 The familiar near-identity formula

When $a = \alpha = 1$, write $h(t) = tu(t)$, so that $f(t) = t + h(t)$. The Euler formula becomes

$$g(y) = y + \sum_{n \geq 1} \frac{(-1)^n}{n!} \left(\frac{d}{dy} \right)^{n-1} h(y)^n. \quad (4.4)$$

Indeed $d^{n-1}(t^n u^n)/dt^{n-1} = t \prod_{j=2}^n (\mathcal{E} + j)u^n$. This short formula already solves both examples. The package adds support management, normalization, exact cutoffs, branch metadata, and independent verification.

A compact Wolfram implementation for a fixed number of correction orders is

```
Clear[UnitInverseJet];
UnitInverseJet[h_, x_Symbol, n_Integer] /; n >= 0 :=
  x + Total[Table[(-1)^k D[h^k, {x, k-1}]/k!, {k, 1, n}]];
Expand[UnitInverseJet[x^2 (1+Log[x]), x, 3]]
```

This expression does not attach an error term and does not enforce a power cutoff. The distinction matters for a mixture of different exponent gaps.

4.3 A coefficient formula on exact sparse blocks

Write $u(t)^n = \sum_{\mu} t^{\mu} Q_{n,\mu}(L)$. The contribution of its (n, μ) block is

$$t^{1+\mu} \frac{(-1)^n}{n! \alpha^n} \prod_{j=1}^{n-1} (\mu + 1 + j\alpha + \partial_L) Q_{n,\mu}(L). \quad (4.5)$$

Only polynomial multiplication and polynomial differentiation are needed. The operators commute and do not raise logarithmic degree.

For a multiindex $m = (m_1, \dots, m_r)$, put $n = |m|$, $\mu = m \cdot \delta$, and $m! = \prod m_j!$. Expanding u^n by the multinomial theorem gives the nonrecursive version

$$g = t + \sum_{m \neq 0} \frac{(-1)^{|m|}}{\alpha^{|m|} m!} t^{1+m \cdot \delta} \prod_{j=1}^{|m|-1} (m \cdot \delta + 1 + j\alpha + \partial_L) \prod_{i=1}^r P_i(L)^{m_i}. \quad (4.6)$$

Different multiindices can give the same exponent. Their polynomial coefficients must be added *before* deciding whether a term vanishes. This is the source of resonances and cancellations even for perfectly ordinary integer exponents.

4.4 Degree bounds and pure-power coefficients

For a fixed multiindex, the logarithmic degree is at most $\sum m_i \deg P_i$. In fact, before cancellations between multiindices, the degree of a nonzero polynomial in (4.6) is unchanged: each factor has a strictly positive constant part $m \cdot \delta + 1 + j\alpha$. In particular, a block of relative weight μ has degree at most

$$d \left\lfloor \frac{\mu}{\delta_*} \right\rfloor. \quad (4.7)$$

This controls coefficient sizes, not analytic tail sizes by itself.

If every $P_i = c_i$ is constant, the coefficient is especially simple:

$$[t^{1+\mu}]g = \sum_{\substack{m \neq 0 \\ m \cdot \delta = \mu}} \frac{(-1)^{|m|}}{\alpha^{|m|} m!} \prod_{j=1}^{|m|-1} (\mu + 1 + j\alpha) \prod_i c_i^{m_i}. \quad (4.8)$$

The sum is finite. No common denominator for the δ_i is needed.

5 Analytic realization and convergence

Theorem 5.1 (Convergence for finite power-logarithmic input). *For (2.1) with finitely many polynomial-logarithmic terms, the series (4.1), evaluated at $t = (y/a)^{1/\alpha}$, converges for all sufficiently small positive y to the inverse in Theorem 2.1. It can be regrouped by increasing exact power weight. If all correction orders $n < N$ are retained, the tail is*

$$\mathcal{O}\left(t^{1+N\delta_*}(1 + |\log t|)^{dN}\right). \quad (5.1)$$

Proof. Write $g = ts$, so the defining equation is (3.4). For each j and $0 \leq k \leq d_j = \deg P_j$, introduce independent small variables

$$z_{j,k} = t^{\delta_j} L^k.$$

The polynomial identity

$$P_j(L + \log s) = \sum_{k=0}^{d_j} \frac{L^k}{k!} P_j^{(k)}(\log s)$$

turns the inverse equation into the finite-dimensional analytic equation

$$\Phi(s, z) = s^\alpha \left[1 + \sum_{j=1}^r s^{\delta_j} \sum_{k=0}^{d_j} \frac{z_{j,k}}{k!} P_j^{(k)}(\log s) \right] - 1 = 0. \quad (5.2)$$

Use the branches of s^q and $\log s$ analytic in a disk about $s = 1$. At $(s, z) = (1, 0)$, we have $\Phi = 0$ and $\partial_s \Phi = \alpha \neq 0$. The analytic implicit-function theorem gives an analytic function $s(z)$ in a fixed polydisk about zero, with $s(0) = 1$.

Every $z_{j,k}(t) \rightarrow 0$ as $t \downarrow 0$. Thus the actual path enters that polydisk and $ts(z(t))$ is the positive inverse. Scaling all the variables simultaneously by λ replaces u by λu . The coefficient of λ^n in $ts(\lambda z(t))$ is exactly the n -th Euler–Lagrange correction. For small t , its analytic radius in λ exceeds one, so evaluation at $\lambda = 1$ is justified.

The multivariate Taylor series is absolutely convergent inside a smaller polydisk. It may therefore be regrouped by total degree and by the locally finite power weights. For a quantitative estimate, choose a closed smaller polydisk on which $|s|$ is bounded, and apply the one-variable Cauchy estimate along λz . With constants independent of small t ,

$$\left| \sum_{n \geq N} [\lambda^n] ts(\lambda z(t)) \right| \leq Ct q(t)^N, \quad q(t) \leq C_1 t^{\delta_*} (1 + |L|)^d < \frac{1}{2}.$$

This proves (5.1). □

Remark 5.2 (What is convergent here). For the logarithmic example, the result is not an ordinary Taylor series in y . It is a convergent regrouping of a multivariate analytic series after substituting small quantities such as y and $y \log y$. The theorem also does not assert that every formally well-based infinite power-log input has a convergent inverse expansion. Finiteness of the analytic lift is an explicit hypothesis.

5.1 Fixed derivatives and analytic remainder control

The analytic lift is also valid in relative complex disks $|t - t_0| < ct_0$, with $c \in (0, 1)$ fixed and $t_0 > 0$ small. A consistent logarithm exists there, and $|t|$ is comparable to t_0 while $|\log t| \leq |\log t_0| + C$. Cauchy estimates on such disks show that a fixed number of derivatives of the degree tail in (5.1) costs the corresponding powers of t_0^{-1} . This is a reason for differentiated remainder estimates in this class. It is *not* a general license to differentiate an arbitrary real-axis big- O statement. The distinction is important when extending the method to tails or residuals known only on the real axis.

6 Exact truncation and the first omitted block

6.1 An exclusive cutoff in the output variable

The package takes a positive real cutoff C and retains precisely the blocks $y^q P(\log y)$ with $q < C$. Since the normalized block $t^{1+\mu}$ becomes $y^{(1+\mu)/\alpha}$ times a constant, the relative cutoff is

$$W = \alpha C - 1. \quad (6.1)$$

If $W \leq 0$, even the leading term is excluded. Otherwise define

$$N = \left\lceil \frac{W}{\delta_*} \right\rceil. \quad (6.2)$$

No correction order $n \geq N$ contributes below the cutoff, since its relative valuation is at least $n\delta_*$. Computing through order N is sufficient to determine both the retained coefficients and a first-omitted boundary block.

An immediate, always valid bound for the truncated inverse G_C is

$$g(y) - G_C(y) = \mathcal{O}\left(y^C (1 + |\log y|)^{dN}\right).$$

That bound is often unnecessarily weak. The implementation returns a sharper power and a sharper logarithmic degree.

6.2 Keeping a boundary while discarding larger powers

At step n , let A_n be the product of the retained part of u^{n-1} with u , before truncation. Split its blocks at relative weight W . The blocks below W are sufficient for the next multiplication. Among all blocks at or above W encountered for $1 \leq n \leq N$, remember the smallest weight μ_0 . Apply the appropriate Euler operator to its coefficient at every order at which it occurs, and add those contributions. Call the resulting polynomial $B(L)$.

This boundary is meaningful even though larger blocks are not propagated. Any descendant of a discarded block has *strictly greater* weight. It cannot return to a smaller exponent or contribute to the smallest omitted weight. Polynomial cancellations inside a product are performed before recording its support.

Theorem 6.1 (First-omitted block and analytic remainder). *Assume $u \neq 0$ and $W > 0$. The boundary procedure above terminates, $W \leq \mu_0 \leq N\delta_*$, and computes the complete coefficient of the inverse at $t^{1+\mu_0}$. With*

$$\rho = \frac{1 + \mu_0}{\alpha}, \quad Q(\ell) = a^{-\rho} B\left(\frac{\ell - \log a}{\alpha}\right),$$

we have

$$g(y) - G_C(y) = y^\rho Q(\log y) + o(y^\rho). \quad (6.3)$$

If $Q \neq 0$, set $k = \deg Q$; otherwise set $k = 0$. In either case,

$$g(y) - G_C(y) = \mathcal{O}\left(y^\rho(1 + |\log y|)^k\right). \quad (6.4)$$

A vanishing Q does not imply exactness.

Proof. The pure least-weight product in u^N has weight $N\delta_*$ and nonzero polynomial coefficient. Its prefixes have weight below W until the first crossing, so some omitted block is encountered by order N . This proves the bound on μ_0 .

A coefficient at μ_0 omitted from an intermediate multiplication could only have an ancestor of weight at least W . Such an ancestor, if nonzero, is itself an encountered omitted block and has weight at least the final minimum μ_0 . Multiplying it by a positive-weight correction would produce weight greater than μ_0 , a contradiction. Thus the complete boundary coefficient is obtained. The same argument proves that all retained coefficients are complete.

In the full formula, the finitely many terms of degrees at most N with weight strictly above μ_0 are $\mathcal{O}(t^{1+\mu_0})$, since a fixed positive power gap dominates every fixed logarithmic power. The remaining degree tail starts at $N + 1$ and is

$$\mathcal{O}\left(t^{1+(N+1)\delta_*}(1 + |L|)^{d(N+1)}\right) = \mathcal{O}(t^{1+\mu_0}).$$

Theorem 5.1 justifies that estimate. Converting from t to y gives (6.3) and then (6.4). $\square$

6.3 Why a plain $\mathcal{O}(y^3)$ would be wrong

For the first example, retaining only $y - y^2(1 + \log y)$ leaves

$$y^3(2\log^2 y + 5\log y + 3) + \mathcal{O}(y^3).$$

After division by y^3 , this is unbounded. The correct simple bound is $\mathcal{O}(y^3(1 + |\log y|)^2)$, not $\mathcal{O}(y^3)$. A formal truncation symbol based only on power valuation must not be read as an analytic Landau estimate without its logarithmic degree.

As another example, the inverse of $x + x^2 + 2x^3$ begins

$$y - y^2 + 0 \cdot y^3 + 5y^4 + \dots$$

At cutoff 3, the boundary exponent is 3 but its polynomial cancels. The package conservatively returns a bound at power 3, with boundary term zero; it does not invent a proof that no later terms exist. Raising the cutoff exposes the next nonzero coefficient.

7 The logarithmic example in full detail

For f_1 , we have $u(t) = t(1 + L)$, $a = \alpha = 1$. Write

$$g_1(y) = y + \sum_{n \geq 1} y^{n+1} P_n(\log y). \quad (7.1)$$

Equation (4.5) gives

$$P_n(L) = \frac{(-1)^n}{n!} \prod_{j=2}^n (n + j + \partial_L)(1 + L)^n. \quad (7.2)$$

Equivalently, the factors are $n + 2 + \partial_L, \dots, 2n + \partial_L$. The empty product for $n = 1$ gives $P_1 = -(1 + L)$.

The first coefficients, independently checked by composition, are

$$\begin{aligned}
P_1(L) &= -1 - L, \\
P_2(L) &= 3 + 5L + 2L^2, \\
P_3(L) &= -\frac{23}{2} - 27L - \frac{41}{2}L^2 - 5L^3, \\
P_4(L) &= \frac{299}{6} + 151L + \frac{335}{2}L^2 + \frac{241}{3}L^3 + 14L^4, \\
P_5(L) &= -\frac{2789}{12} - \frac{5177}{6}L - 1256L^2 - \frac{5365}{6}L^3 - \frac{3727}{12}L^4 - 42L^5.
\end{aligned} \tag{7.3}$$

Thus an answer through the fourth power of y is

$$\begin{aligned}
g_1(y) &= y - y^2(1 + L) + y^3(3 + 5L + 2L^2) \\
&\quad - y^4 \left(\frac{23}{2} + 27L + \frac{41}{2}L^2 + 5L^3 \right) + \mathcal{O}(y^5(1 + |L|)^4), \quad L = \log y.
\end{aligned} \tag{7.4}$$

7.1 An elementary triangular check

Put $g = ys$, $s = 1 + c_1y + c_2y^2 + \dots$. The equation becomes

$$s + ys^2(1 + \log y + \log s) = 1.$$

At first order, $c_1 = -(1 + L)$. At second order,

$$c_2 + (3 + 2L)c_1 = 0, \quad c_2 = (1 + L)(3 + 2L).$$

Continuing gives (7.3). This calculation shows explicitly where the derivative of the logarithm enters: the $\log s$ term contributes at the next order and cannot be omitted. The independent validation script solves this triangular system through three corrections.

7.2 Catalan numbers in the top logarithmic degree

Every factor in (7.2) preserves degree and multiplies the leading coefficient by its constant part. Therefore

$$\deg P_n = n, \quad [L^n]P_n(L) = (-1)^n \frac{(2n)!}{n!(n+1)!} = (-1)^n \text{Cat}_n. \tag{7.5}$$

The sequence $-1, 2, -5, 14, -42, 132, \dots$ in the leading logarithms is therefore forced by the universal formula. It also follows from keeping only the dominant $y \log y$ perturbation in the analytic lift. The lower logarithmic coefficients encode the interactions with the constant part of $1 + \log y$ and with $\log s$.

7.3 A two-variable convergence picture

For this example, a particularly economical analytic lift is

$$s + ws^2 + zs^2 \log s = 1, \quad z = y, \quad w = y(1 + \log y).$$

The left side has derivative one with respect to s at $(s, z, w) = (1, 0, 0)$. Its solution is analytic in (z, w) near the origin. The path $(y, y(1 + \log y))$ tends to that origin, so the power-log expansion converges for sufficiently small positive y . This is an explicit resolution of the apparent tension between the logarithmic endpoint singularity and the convergence theorem.

8 The irrational-power example and a closed coefficient law

Let $p > 1$ be any fixed real number and consider $f_p(x) = x + x^p$. The Euler formula gives

$$g_p(y) = \sum_{n \geq 0} b_n(p) y^{1+n(p-1)}, \quad b_n(p) = \frac{(-1)^n}{1+n(p-1)} \binom{np}{n}. \quad (8.1)$$

For $n \geq 1$, an equivalent finite-product expression is

$$b_n(p) = \frac{(-1)^n}{n!} (np)^{n-1} = \frac{(-1)^n \Gamma(np+1)}{\Gamma(n+1) \Gamma(n(p-1)+2)}. \quad (8.2)$$

The finite product is convenient for exact algebra; the gamma expression is convenient for coefficient asymptotics. The first terms are

$$\begin{aligned} g_p(y) = & y - y^p + p y^{2p-1} - \frac{p(3p-1)}{2} y^{3p-2} \\ & + \frac{p(4p-1)(4p-2)}{6} y^{4p-3} - \frac{p(5p-1)(5p-2)(5p-3)}{24} y^{5p-4} + \dots \end{aligned} \quad (8.3)$$

For $p = \sqrt{2}$, this yields exactly the requested beginning:

$$\begin{aligned} g_2(y) = & y - y^{\sqrt{2}} + \sqrt{2} y^{2\sqrt{2}-1} - \frac{6-\sqrt{2}}{2} y^{3\sqrt{2}-2} \\ & + \left(\frac{17\sqrt{2}}{3} - 4 \right) y^{4\sqrt{2}-3} + \mathcal{O}(y^{5\sqrt{2}-4}). \end{aligned} \quad (8.4)$$

Here the error has no logarithmic factor.

8.1 An ordinary analytic series in a nonordinary variable

Set $z = y^{p-1}$ and $g_p(y) = yH(z)$. The inverse equation is

$$H + zH^p = 1, \quad H(0) = 1. \quad (8.5)$$

This is an ordinary analytic implicit equation near $(H, z) = (1, 0)$, even when p is irrational. Its ordinary Taylor coefficients in z are the $b_n(p)$. The irrational exponents appear only when substituting $z = y^{p-1}$ back into the result.

8.2 Radius of convergence

Stirling's formula applied to (8.2) gives, with $\delta = p-1$,

$$|b_n(p)| \sim \frac{\sqrt{p}}{\sqrt{2\pi} \delta^{3/2}} n^{-3/2} \left(\frac{p^p}{\delta^\delta} \right)^n. \quad (8.6)$$

Hence the exact Taylor radius in z is

$$R_p = \frac{(p-1)^{p-1}}{p^p}. \quad (8.7)$$

The critical point of (8.5) is consistent with this: $H_* = p/(p-1)$ and $z_* = -R_p$, where $1 + pzH^{p-1} = 0$. For $p = \sqrt{2}$,

$$R_p \approx 0.4251956066518408, \quad R_p^{1/(p-1)} \approx 0.1268627051233097.$$

Thus the inverse series converges, in particular, for $0 < y < R_p^{1/(p-1)}$. The positive inverse itself exists beyond this interval; the bound describes the Taylor series in z , not the global domain of the inverse. The $n^{-3/2}$ factor also gives absolute convergence on the circle $|z| = R_p$, although the negative critical point is singular.

9 Mixed exponents, scaling, and resonances

9.1 A leading monomial other than x

For

$$f(x) = 4x^2(1 + x(1 + \log x)), \quad t = \frac{\sqrt{y}}{2}, \quad L = \log t,$$

the first two correction orders from (4.1) are

$$g(y) = t - \frac{1}{2}t^2(1 + L) + \frac{1}{8}t^3(7 + 12L + 5L^2) + \mathcal{O}(t^4(1 + |L|)^3). \quad (9.1)$$

The coefficient $7 + 12L + 5L^2$ comes from $(5 + \partial_L)(1 + L)^2$. In terms of y , even the first correction contains the shift $\log t = \frac{1}{2} \log y - \log 2$. A normalization that keeps $\log y$ unchanged would give wrong coefficients.

More generally, one relative correction $x^\delta P(\log x)$ gives

$$\begin{aligned} g(y) = t - \frac{t^{1+\delta}}{\alpha} P(L) \\ + \frac{t^{1+2\delta}}{2\alpha^2} [(2\delta + 1 + \alpha)P(L)^2 + 2P(L)P'(L)] + \dots \end{aligned} \quad (9.2)$$

This is a useful unit test for both the exponent rescaling and the logarithm shift.

9.2 Two generators

For $u(t) = c_1 t^{\delta_1} + c_2 t^{\delta_2}$ with $\alpha = 1$, the first mixed correction, from $(m_1, m_2) = (1, 1)$, is

$$(2 + \delta_1 + \delta_2)c_1 c_2 t^{1+\delta_1+\delta_2}.$$

When the ratio of the generators is irrational, most weights are distinct; when it is rational, multiple multiindices can meet at the same weight. Either case is handled by the same sparse addition operation.

For example, the source $x + x^{\sqrt{2}} + x^{\sqrt{3}}$ has relative generators $\sqrt{2} - 1$ and $\sqrt{3} - 1$. The package orders the resulting real algebraic exponents exactly. The validation includes independent composition for this genuinely two-generator example, as well as for $x + x^{\sqrt{2}} + x^2(1 + \log x)$.

9.3 Collision handling is a correctness condition

For $f = x + x^2 + 2x^3$, the contribution at inverse power y^3 from the linear perturbation order is $-2y^3$, whereas the quadratic perturbation order contributes $+2y^3$. Recording the two terms under different unevaluated exponent keys or deciding their order numerically could prevent the cancellation. The implementation compares mathematical exponents and adds their full coefficient polynomials before testing for zero.

10 Residuals, source uncertainty, and conditioning

10.1 What an independent residual check should verify

Let $\beta = 1/\alpha$, and suppose an inverse jet has error $g - G = \mathcal{O}(y^\rho(1 + |\log y|)^k)$, with $G \sim g \sim t$. The mean-value theorem and the derivative equivalent in Theorem 2.1 give

$$f(G(y)) - y = \mathcal{O}\left(y^{\rho+1-\beta}(1 + |\log y|)^k\right). \quad (10.1)$$

Thus the symbolic checker should find no residual block with power strictly below

$$C_{\text{res}} = \rho + 1 - \beta. \quad (10.2)$$

Comparing the residual and inverse errors at the same power would be wrong when $\alpha \neq 1$.

The package checker recomputes composition using the binomial and logarithm identities (3.2). For an inner jet $G = cy^\beta(1+v)$, a source block is composed as

$$G^q P(\log G) = c^q y^{\beta q} (1+v)^q P(\log c + \beta \log y + \log(1+v)). \quad (10.3)$$

This algorithm does not call the Euler coefficient generator. Passing the check is an independent algebraic consistency test. It does not by itself prove a global branch statement, establish the assumptions supplied by the caller, or certify a floating-point error bound.

10.2 Transporting an asymptotic input remainder

Theorem 10.1 (Input-jet error transport). *Let f_0 belong to (2.1). Suppose $f = f_0 + R$ is continuous, strictly increasing near zero, and*

$$R(x) = \mathcal{O}(x^{\alpha+\sigma}(1 + |\log x|)^m), \quad \sigma > 0.$$

Then the positive inverse branches satisfy

$$f^{-1}(y) - f_0^{-1}(y) = \mathcal{O}\left(y^{(1+\sigma)/\alpha}(1 + |\log y|)^m\right). \quad (10.4)$$

Proof. Both sources have the same leading equivalent ax^α , so both inverses are asymptotic to $t = (y/a)^{1/\alpha}$. If $g = f^{-1}(y)$ and $g_0 = f_0^{-1}(y)$, then $f_0(g) - f_0(g_0) = -R(g)$. The mean-value theorem for the known smooth source f_0 divides this error by a quantity comparable to $t^{\alpha-1}$. The result is $\mathcal{O}(t^{1+\sigma}(1 + |\log t|)^m)$, which is (10.4). No derivative bound on R is needed once the stated monotonicity and existence of the branch for f are available. $\square$

For example, replacing $\sin x$ by $x - x^3/6 + x^5/120$ has a source error $\mathcal{O}(x^7)$. Its inverse coefficients below power y^7 agree with those of $\arcsin y$, but a boundary coefficient at y^7 computed from that finite polynomial is not necessarily the boundary coefficient of the original inverse. The package assumes that its supplied finite expression is the exact source. A caller using an asymptotic source jet must enforce this independent error budget; there is no implicit claim of infinite input precision.

10.3 A posteriori numerical bounds

For a positive approximation G , if $f' \geq m_0 > 0$ on an interval containing G and the desired root, then

$$|g(y) - G(y)| \leq \frac{|f(G(y)) - y|}{m_0}. \quad (10.5)$$

For the two original examples, global slope bounds give particularly simple versions:

$$|g_1 - G| \leq \frac{|f_1(G) - y|}{1 - 2e^{-5/2}}, \quad |g_p - G| \leq |f_p(G) - y| \quad (p > 1),$$

provided $G > 0$. These are mathematical inequalities. A numerically *certified* evaluation additionally requires a rigorous enclosure of the residual; ordinary high-precision arithmetic is evidence, not an interval certificate.

The relative condition number tends to

$$\left| \frac{yg'(y)}{g(y)} \right| \longrightarrow \frac{1}{\alpha}.$$

This clean normalization explains why working with g/t , rather than with a poorly scaled raw implicit equation, is helpful for both formal and numerical algorithms.

11 The Wolfram Language package

11.1 Loading and the two original examples

The implementation is in `Kernel/RealInverseAsymptotics.wl`. It uses no external Wolfram packages. Load the file and keep the source and output variables distinct and unassigned:

```
Get["/path/to/RealInverseAsymptotics/Kernel/RealInverseAsymptotics.wl"];
Clear[x, y];

r1 = RealInverseAsymptotic[
  x + x^2 (1 + Log[x]), {x, 0}, {y, 4}];
r1["Expression"]
(* y - y^2 (1+Log[y]) + y^3 (3+5 Log[y]+2 Log[y]^2) *)
r1["Remainder"]
(* PowerLogO[y, 4, 3] *)

r2 = RealInverseAsymptotic[
  x + x^Sqrt[2], {x, 0}, {y, 4 Sqrt[2]-3}];
r2["Expression"]
(* y-y^Sqrt[2]+Sqrt[2] y^(2 Sqrt[2]-1)
  -(6-Sqrt[2])/2 y^(3 Sqrt[2]-2) *)
```

The comments show mathematically verified expected expressions, not a transcript from an executed Wolfram kernel. This distinction applies to all package examples in this article.

The first call retains powers strictly below 4, not four terms and not all powers through 4 inclusively. The second cutoff is chosen to exclude the next power $4\sqrt{2} - 3$, so it returns the four blocks requested in the question. A cutoff such as 3 is also accepted; its meaning is the same exclusive real-exponent bound.

11.2 The result object

The function returns a standard `Association`. Its main fields are listed in Table 1. Keeping the finite expression separate from the remainder avoids pretending that a generalized series belongs to `SeriesData`.

Field	Meaning
Expression	The retained finite inverse expression.
Terms	Sorted pairs $\{q, P(\log y)\}$, one per exact power.
RemainderPower	The first possible omitted output exponent ρ .
RemainderLogDegree	The nonnegative integer k in the returned bound.
RemainderScale	$y^\rho(1 + \log y)^k$, or zero for an exact result.
Remainder	The inert annotation <code>PowerLogO[y, rho, k]</code> .
BoundaryTerm	The complete term $y^\rho Q(\log y)$; it can cancel.
LeadingTerm	$(y/a)^{1/\alpha}$, the selected branch scale.
Exact	True only for an untruncated pure-monomial source.
Cutoff	The requested exclusive exponent bound.

Table 1: Principal output fields. Source variables, source expression, assumptions, branch information, and the finite order bound are also stored.

`PowerLogO` has no arithmetic rules. It is a declaration of an asymptotic bound, not an object that can safely be substituted, multiplied, or differentiated without additional analysis. To evaluate numerically, use `r["Expression"]` and only then apply `N`; do not evaluate the exponents approximately before constructing the support.

If the cutoff excludes the leading term, the retained expression is zero and the omitted boundary is the leading term. The residual checker refuses such an empty inverse approximation because composition through its logarithm would be undefined. For a pure source ax^α whose leading term is retained, the inverse is exact and the remainder is zero.

11.3 Parser and coefficient normalization

The parser first replaces $\text{Log}[x]$ by a fresh polynomial coordinate, then uses ordinary `Expand`. Each resulting summand must be a product of a power x^q and an x -independent polynomial in that coordinate. Equal exponents are merged. The smallest exponent must be a positive constant α , and its coefficient must be an L -independent constant proved positive under the supplied assumptions. Dividing the remaining blocks by ax^α constructs the positive correction gaps.

The parser deliberately does not perform branch-sensitive rewrites. For example, normalize $\log(2x)$ to $\log 2 + \log x$ first on the positive domain; an unreduced $\text{Log}[2x]$ is rejected. Noninteger powers of products are not distributed using `PowerExpand`. The only power and logarithm transformations used by the algorithm are those justified by the explicit positive leading coefficient and the positive local branch.

All real-number literals with finite precision are rejected, even in coefficients or the cutoff. Inputs such as `Sqrt[2]` and `7/5` are exact; `1.41421356` and `1.4` are not. Symbolic coefficients are allowed when their reality and the positivity of the leading coefficient can be established, for example

```
RealInverseAsymptotic[
  a x + a x^2, {x, 0}, {y, 3}, Assumptions -> a > 0]
```

Exponents, by contrast, must be numerically specified exact real constants. An unspecialized symbol p , even with $p > 1$, is rejected. A symbolic exponent could change the ordering, collision pattern, and number of retained terms as a parameter varies; a correct fully symbolic implementation would need a parameter-space case decomposition. The present implementation does not silently choose one such case.

11.4 Exact exponent equality and order

Each exponent is simplified exactly. Comparisons must resolve to less than, equal to, or greater than; otherwise the computation returns `Failure["UndecidableExponentOrder", ...]`. There is no explicit floating-point fallback, tolerance-based equality, or machine-number hash key. Real algebraic expressions, including the original $\sqrt{2}$, are the primary intended case. Exact transcendental constants may also work when all necessary inequalities and iteration bounds are resolved.

The reference implementation uses sorted lists and equality-checked merging rather than a sophisticated algebraic-number key. This is deliberately conservative about correctness. It is not a claim that list insertion is an optimal data structure for very large supports.

11.5 The main loop

The computational core implements the following finite procedure.

1. Parse and normalize f ; compute $W = \alpha C - 1$, δ_* , and $N = \lceil W/\delta_* \rceil$.
2. Start with the zeroth power $u^0 = 1$, retained inverse coefficient one, and no omitted boundary.
3. For $n = 1, \dots, N$, multiply the retained part of u^{n-1} by u . For each raw block, apply (4.5). Add blocks below W to the inverse and retain their untransformed product coefficients for the

next multiplication. For omitted blocks, update the smallest boundary weight and its summed transformed coefficient.

4. Convert $t^{1+\mu}Q(L)$ to $y^{(1+\mu)/\alpha}a^{-(1+\mu)/\alpha}Q((\log y - \log a)/\alpha)$. Return the expression and the boundary-based remainder data.

The distinction between the untransformed product coefficient and the Euler-transformed inverse coefficient is essential: the next iteration needs u^n , not the n -th inverse correction.

The operator on one block is implemented by the short loop

```
eulerBlock[q_, mu_, alpha_, n_Integer, ell_Symbol] :=
Module[{r = q, j},
  Do[r = poly[(mu + 1 + j alpha) r + D[r, ell]],
    {j, 1, n - 1}];
  poly[(-1)^n r/(Factorial[n] alpha^n)]
];
```

Here `poly` expands and simplifies polynomial coefficients under the active assumptions. The surrounding package supplies parsing, merging, cutoff handling, and failures.

11.6 Independent composition and the checker

Two additional public functions are supplied:

```
PowerLogCompose[Log[x], x, y+y^2, {y,4}]
(* Expression: Log[y]+y-y^2/2+y^3/3 *)

CheckInverseAsymptotic[r1]
(* Passed -> True is the expected result. *)
```

The composition engine implements (10.3), using finite binomial and logarithm sums cut off by their positive valuations. It does not reuse the inverse coefficient formula. `CheckInverseAsymptotic` uses the source expression saved in the result and the residual cutoff (10.2). Its output includes the Boolean `Passed`, the residual expression below that bound, and the bound itself. A test deliberately corrupts an inverse coefficient to ensure that the checker is intended to reject an incorrect jet.

11.7 Budgets, failure behavior, and cost

The options are

```
Assumptions :> $Assumptions
"MaxOrder" -> 256
"MaxTerms" -> 20000
```

A temporary multiplication budget is four times `MaxTerms`. Exceeding a budget returns a structured failure rather than a silently incomplete series. These are size budgets, not a guarantee of a wall-clock running-time bound for symbolic simplification.

If M is a bound on retained support size and the source has r blocks, a multiplication produces at most rM raw candidates before merging. The straightforward equality-checked merge can take quadratic time in that candidate count. Polynomial degrees grow at most linearly with perturbation order by (4.7); coefficient bit sizes and exact algebraic comparison costs are additional factors. The implementation is appropriate for moderate symbolic orders and for auditing the mathematics. At much higher orders, a better support index and Newton reversion are natural optimizations, but their complexity is not

claimed for this implementation.

12 Validation, reproducibility, and what was not executed

12.1 Independent computations that were executed

The script `validation/validate.py` completed 58 checks using SymPy 1.14.0 and mpmath 1.3.0. Its JSON report is included in the archive. The exact checks cover the first logarithmic coefficients, the general irrational Fuss–Catalan formula through nine correction indices, independent composition, a separate triangular coefficient solve, exact monomial normalization, strict cutoff edge cases, and a cancelled boundary.

Nine additional source classes were checked by independent composition: a rational nonunit leading exponent; a nonunit leading coefficient; resonant integer exponents; mixed logarithmic polynomials; a negative perturbation; an irrational leading exponent; two distinct irrational generators; an irrational/logarithmic mixture; and a polynomial logarithmic coefficient with several signs. These tests exercise the normalization and support logic, not only the two motivating examples.

Numerical checks used 100-digit arithmetic and positive bisection to calculate an inverse independently of the coefficient formula. For each original example, the script compares approximations with one, two, three, and five correction orders at $y = 10^{-3}, 10^{-6}, 10^{-12}$. Table 2 shows selected values. The ratio compares the actual absolute error with the absolute value of the computed boundary term; its approach to one is consistent with (6.3).

Source	y	Orders	Absolute error	Error / boundary
Logarithmic	10^{-3}	3	$1.2594119 \cdot 10^{-11}$	1.01590491
Logarithmic	10^{-6}	3	$3.2814172 \cdot 10^{-25}$	1.00003643
Logarithmic	10^{-12}	3	$6.5895650 \cdot 10^{-54}$	1.00000000
Logarithmic	10^{-6}	5	$4.5579380 \cdot 10^{-34}$	1.00003940
Irrational	10^{-3}	3	$3.8876652 \cdot 10^{-8}$	0.90508123
Irrational	10^{-6}	3	$4.5691233 \cdot 10^{-16}$	0.99401819
Irrational	10^{-12}	3	$5.2638620 \cdot 10^{-32}$	0.99998031
Irrational	10^{-6}	5	$1.7139602 \cdot 10^{-20}$	0.99364138

Table 2: Executed high-precision numerical checks. “Orders” counts correction orders after the leading term. Values are numerical evidence, not interval-arithmetic certificates. Full data are in `validation/numerics.csv`.

12.2 Wolfram tests supplied for execution

The archive contains 35 `MUnit VerificationTest` cases. They check both original examples, the logarithmic remainder, the complete omitted polynomial, a nonunit leading monomial, an exact monomial, exponent collisions and cancellation, symbolic positive coefficients, independent composition, a deliberately corrupted result, and error handling. They also specify expected rejection of inexact data, an unspecialized symbolic exponent, a logarithmic leading scale, an essential exponential, complex coefficients, and insufficient resource budgets.

Run the suite inside Mathematica with

```
TestReport["/path/to/RealInverseAsymptotics/Tests/RealInverseAsymptotics.wlt"]
```

or from a command line with

```
wolframscript -file Tests/RunTests.wls
```

These Wolfram tests were not executed in this session. The available Wolfram connector returned a tool-availability error. No local Wolfram kernel was installed, and an attempted Mathics installation was unavailable. Consequently there is no claimed Wolfram pass count, no verified current-kernel compatibility claim, and no claim to have reproduced the old `InverseFunction` behavior on a current release. Independent SymPy/mpmath execution validates the derived formulas and an independently coded algorithm; it does not execute Wolfram evaluation semantics. A separate delimiter/comment/string balance check of the supplied Wolfram sources is included, but it is not a Wolfram parser or runtime test.

12.3 Reproducing the independent checks and the document

The validation scripts and raw results are included rather than only a summary. Their commands are

```
python validation/validate.py
python validation/check_wl_delimiters.py
```

The article source is self-contained apart from standard LaTeX packages. From the `article` directory, compile it with

```
latexmk -pdf real_inverse_asymptotics.tex
```

The package and the accompanying original material in this archive are provided under the MIT No Attribution license. The cited literature remains under its respective terms and is not bundled as third-party full text.

13 Extensions, alternative algorithms, and scope boundaries

13.1 Newton reversion in the same algebra

The normalized equation $\Phi_t(s) = s^\alpha(1 + u(ts)) - 1 = 0$ has derivative with leading constant α . Newton iteration therefore has the formal form

$$s_{k+1} = s_k - \frac{s_k^\alpha(1 + u(ts_k)) - 1}{\alpha s_k^{\alpha-1}(1 + u(ts_k)) + t s_k^\alpha u'(ts_k)}. \quad (13.1)$$

Starting from $s_0 = 1$, a Taylor expansion of Φ_t shows that an error of positive valuation η is replaced by one of valuation at least 2η . Its formal derivative is a unit, and the quadratic remainder contains two factors of the previous error. Thus relative accuracy at least $2^k \delta_*$ is available after k Newton steps, when all arithmetic is performed to the needed guard precision.

This gives an alternative algorithm, not an additional implemented backend. It uses composition, reciprocal, and generalized unit powers; all are available mathematically in the same algebra. The Euler formula remains particularly useful for direct coefficients, closed forms, degree laws, and independent checks. A future high-order implementation should state its guard-precision rules explicitly rather than assuming that a finite jet may be differentiated or composed without changing its error budget.

13.2 Observables of the inverse

The parameter identity (4.2) also expands an observable $\Psi(g)$. For a fixed real exponent q , it gives

$$g(y)^q = t^q \left[1 + \sum_{n \geq 1} \frac{(-1)^n q}{n! \alpha^n} \prod_{j=1}^{n-1} (\mathcal{E} + q + j\alpha) u(t)^n \right]. \quad (13.2)$$

For $q = 0$ the correction vanishes. Differentiating with respect to q at zero gives

$$\log g(y) = \log t + \sum_{n \geq 1} \frac{(-1)^n}{n! \alpha^n} \prod_{j=1}^{n-1} (\mathcal{E} + j\alpha) u(t)^n. \quad (13.3)$$

These are useful theoretical formulas for applications that need a power, logarithm, or derivative of the inverse rather than the inverse alone. The shipped package obtains such finite jets through `PowerLogCompose`; it does not expose separate optimized generators for (13.2) and (13.3).

13.3 Other endpoints and branches

A finite basepoint can be treated by introducing positive local coordinates. For a right-hand branch through x_0 , write $x = x_0 + t$ and subtract the image value $f(x_0)$. For a left-hand branch, use $x = x_0 - t$ and account for the sign of the output coordinate. The resulting positive-coordinate source must then satisfy the input contract. This transformation is an explicit caller operation; the package currently accepts the basepoint zero only.

An asymptotic inverse at infinity can sometimes be converted to one at zero by $x = 1/t$ and an appropriate reciprocal output variable. Whether the transformed source lies in the finite power-log class has to be checked. A change of variables is not a reason to discard a leading logarithmic or exponential scale that the coefficient algebra cannot represent.

13.4 A logarithmic leading scale requires a different normalization

A source such as

$$f(x) = ax^\alpha (-\log x)^b$$

is outside (2.1) when $b \neq 0$. Its leading coefficient is not constant and its inverse is not asymptotic to $(y/a)^{1/\alpha}$. There is nevertheless a useful explicit leading-scale inversion. Put $\ell = -\log x$. For $b \neq 0$,

$$\ell = -\frac{b}{\alpha} W_k \left(-\frac{\alpha}{b} (y/a)^{1/b} \right), \quad x = e^{-\ell}. \quad (13.4)$$

For $b > 0$ and $y \downarrow 0$, the required large positive ℓ uses W_{-1} ; for $b < 0$, it uses W_0 on a large positive argument. This follows directly by rewriting $(y/a)^{1/b} = \ell e^{-\alpha\ell/b}$.

For example, the small positive inverse of $x(-\log x)$ is

$$x = -\frac{y}{W_{-1}(-y)} \sim \frac{y}{\log(1/y)}.$$

Its corrections involve $\log \log(1/y)$. They require a nested logarithmic scale or an exact leading Lambert expression. The current package rejects this input rather than misclassifying a zero correction gap as positive. Formula (13.4) describes a possible normalization adapter, not an implemented universal log-leading solver.

13.5 Exponentially small sectors and Fabius-type applications

For $f(x) = e^{-1/x^\beta}$, the inverse is $(\log(1/y))^{-1/\beta}$. Its source has no positive power leading monomial at zero. For $f(x) = x + e^{-1/x}$, the power-log inverse jet is simply y to every algebraic order, but the inverse is not exactly y : formally,

$$g(y) = y - e^{-1/y} + y^{-2}e^{-2/y} + \dots$$

This illustrates why an all-orders power-log jet does not determine exponentially small corrections. The sparse package rejects the exponential input; it does not silently replace it by its power-log projection.

For the supplied Fabius research context, an appropriate larger architecture would first identify a leading tail scale, transform to a coordinate where the inverse equation is near identity, then state the support and derivative bounds of every residual retained in that coordinate. Only the resulting power-logarithmic subproblem, when it satisfies (2.1), can be handed to this package unchanged. A real-axis tail equivalent alone is not a proof of a complex collar, of differentiated residual bounds, or of the admissibility of a chosen transseries substitution. Those distinctions are exactly the sort of support and realization issues emphasized in the supplied notes [2].

13.6 The precise boundary of this implementation

The implemented class includes finite sums of exact real powers with polynomial logarithmic coefficients, positive monomial leading terms, several irrational generators, and real symbolic coefficients under sufficient assumptions. It excludes a logarithmic leading monomial, negative powers of the logarithm, iterated logarithms, essential exponentials, oscillatory coefficients, arbitrary implicit equations, and unspecialized symbolic exponents. Some excluded problems are easier than the included ones after a suitable transformation; exclusion here is an interface and proof-contract boundary, not a mathematical impossibility.

14 Conclusions

The two motivating examples do not require a closed form for the inverse or an ordinary Taylor series at zero. They require an explicitly selected positive germ, a coefficient algebra that retains logarithmic differentiation, and an exact real-exponent support.

For a finite power-logarithmic perturbation of ax^α , these needs are met by the Euler–Lagrange formula (4.1). Its support is a locally finite positive semigroup. The analytic lift proves convergence for small positive arguments, and the first-omitted theorem turns formal truncation into an honest logarithmic Landau bound. The sparse package implements this finite construction and independently checks the resulting coefficients by composition. The two original inverse expansions follow as direct specializations, while the normalization and error-transport results make the method usable for a substantially broader family of real inverse problems.

A Additional coefficient and implementation checks

The sixth logarithmic polynomial, useful for checking the first omitted term when retaining five corrections, is

$$\begin{aligned} P_6(L) = & 132L^6 + \frac{5981}{5}L^5 + \frac{52937}{12}L^4 + \frac{16979}{2}L^3 \\ & + 9007L^2 + \frac{30019}{6}L + \frac{22769}{20}. \end{aligned}$$

Its leading coefficient is the sixth Catalan number, consistent with (7.5). The validation report contains this polynomial and the raw numerical comparisons that use it as an omitted boundary term.

For a one-block source, the general formula also makes hand checks easy. If $u = t^\delta P(L)$, the first three relative coefficients are

$$\begin{aligned} A_1 &= -\frac{P}{\alpha}, \\ A_2 &= \frac{1}{2\alpha^2}(2\delta + 1 + \alpha + \partial_L)P^2, \\ A_3 &= -\frac{1}{6\alpha^3}(3\delta + 1 + \alpha + \partial_L)(3\delta + 1 + 2\alpha + \partial_L)P^3. \end{aligned}$$

They multiply $t^{1+\delta}$, $t^{1+2\delta}$, and $t^{1+3\delta}$, respectively. Setting $\alpha = 1$, $\delta = p - 1$, and $P = 1$ recovers the irrational-power coefficients; setting $\delta = 1$ and $P = 1 + L$ recovers the logarithmic example.

B Archive map

```
RealInverseAsymptotics/
  README.md
  LICENSE
  Kernel/
    RealInverseAsymptotics.wl
    init.m
  Examples/
    Examples.wl
  Tests/
    RealInverseAsymptotics.wlt
    RunTests.wls
  article/
    real_inverse_asymptotics.tex
    real_inverse_asymptotics.pdf
  validation/
    validate.py
    validation_report.json
    numerics.csv
    example_coefficients.tex
    run.log
    check_wl_delimiters.py
    wl_delimiter_report.json
```

References

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